GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Point Machine Motor Inspection and Trend Log

For S&T maintainers and signalling contractors working with DC series-wound point machine motors

Use this log at commissioning and at every scheduled inspection. Acceptance limits come from the machine's RDSO specification, the Signal Engineering Manual and your zonal maintenance schedule; this document sets out what to measure and how to interpret changes against the baseline for each point.

Section 1 — Safety and preparation

- ☐ Work only as authorised S&T staff under the disconnection, reconnection and testing procedures of the Signal Engineering Manual and zonal instructions.
- ☐ Record point number, machine type, motor variant (110 V or 120 V DC), gear ratio and AC immunity rating.
- ☐ Record weather, ambient temperature and whether the track was recently water-logged.
- ☐ Have ready: DC clamp meter or shunt, true-RMS multimeter, 500 V megger, stopwatch or data logger and brush gauge.

Section 2 — Track and mechanical checks first

- ☐ Inspect for obstruction between tongue rail and stock rail.
- ☐ Check that slide chairs are clean and lubricated.
- ☐ Check packing, gauge and rail creep at the switch.
- ☐ Check detection rods, lock rods, stretcher bars and clamp lock (where fitted) against the drawing.
- ☐ Carry out the obstruction test with the specified test piece and confirm the point does not lock or prove detection.
- ☐ If the point throws fully but shows failed, check detection contacts and wiring.

Section 3 — Measurements during a throw (normal and reverse)

- ☐ Working current: steady value and peak, in each direction.
- ☐ Throw time from operation to detection, in each direction.
- ☐ Voltage at the motor terminals during the throw, and at the feed end.
- ☐ Clutch-slip current during the obstruction test; compare with the machine's specified value.
- ☐ Compare with the baseline: rising current → growing load; longer throw at normal current → low terminal voltage.
- ☐ During repeated operations allow cooling time and respect the motor's 10-minute rating.

Section 4 — Insulation resistance

- ☐ Test winding to frame with a 500 V megger after safe disconnection.
- ☐ Test feed cable cores to earth separately from the motor.
- ☐ Record motor temperature and humidity with each reading.
- ☐ Judge against the zonal schedule limit and the trend for that motor.
- ☐ After water-logging, retest before returning to service; dry or replace the motor if readings have fallen.

Section 5 — Brushes, commutator and gears

- [ ] Measure brush length against the manufacturer's minimum mark.
- [ ] Check spring pressure and free movement of brushes in their holders.
- [ ] Inspect the commutator: uniform brown film is healthy; grooves, burnt bars or carbon tracking are not.
- [ ] Remove carbon dust from inside the motor housing.
- [ ] Check gear lubrication, backlash and lost motion.
- [ ] Check seals, gaskets and cable glands for water entry paths.

Section 6 — When replacing a motor

- [ ] Match the voltage variant (110 V or 120 V DC) and gear ratio (20:1 or 27:1).
- [ ] Match the AC immunity variant (160 V AC or 400 V AC) to the section's requirement.
- [ ] Consider the watertight IP67 motor where the location is prone to water-logging.
- [ ] Record new baseline readings in both directions before handing the point back.

Trend log (one sheet per point machine)

Parameter	Unit	Baseline	This visit	Reference
Working current, normal	A	_____	_____	Baseline; 5.3 A max rating
Working current, reverse	A	_____	_____	Baseline; 5.3 A max rating
Throw time, normal	s	_____	_____	Machine specification
Throw time, reverse	s	_____	_____	Machine specification
Terminal voltage during throw	V	_____	_____	Feed design
Clutch-slip current	A	_____	_____	RDSO specification
IR, winding to frame (500 V)	MΩ	_____	_____	Zonal schedule
IR, cable cores to earth	MΩ	_____	_____	Zonal schedule
Brush length	mm	_____	_____	Manufacturer minimum

This log does not set acceptance limits. Always apply the limits in the machine's RDSO specification, the Signal Engineering Manual and your zonal maintenance schedule.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.